



Beyond conventional antibiotics approaches: Global perspectives on alternative therapeutics including herbal prevention, and proactive management strategies in bovine mastitis

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ABSTRACT

Mastitis, an intramammary inflammation resulting from microbial infectious agents, continues to pose a significant challenge within the dairy sector, adversely affecting animal well-being and leading to substantial economic losses. These losses are attributed to decreased milk production, heightened culling rates, and the expenses related to diagnostics, veterinary care, medication, and labor. Moreover, additional costs emerge due to reduced forthcoming milk yields, compromised reproductive health, and increased susceptibility to various illnesses. Identifying the responsible agents is crucial for disease management and the implementation of antimicrobial treatments. Despite the prevalent use of antibiotic treatment, the pressing need for new therapeutic alternatives to combat bovine mastitis arises from limitations, including low cure rates, rising resistance, and the presence of antibiotic residues in milk. This review explores the potential application of herbal extracts and essential oils known for their antimicrobial properties as alternative options for managing pathogens in mastitis treatment. It examines various treatment methods and management strategies, particularly emphasizing the progress of herbal remedies and natural therapeutics in addressing mastitis, a significant concern in bovine populations and dairy herds.

1. Introduction

Mastitis, a bovine affliction, exerts substantial detrimental effects on the agricultural economy and the well-being of livestock [1]. Bovine mastitis, characterized by udder and mammary tissue inflammation due to microbial infections or environmental stress, substantially reduces dairy industry production and negatively impacts global animal farming profitability [2]. The Greek terms “mastos” and “itis,” which mean “breast” and “inflammation,” respectively, are the source of the English word “mastitis” [3]. Bovine mastitis is divided into three categories: clinical, subclinical, and chronic, depending on the severity of the illness [4]. Visible symptoms including erythematous and swollen mammary glands, elevated body temperature, and the presence of thin, watery milk with flakes and clots facilitate the straightforward identification of clinical mastitis [5]. Scientists report that severe clinical mastitis caused by *E. coli* can lead to irreversible damage to mammary tissue, total loss of

milk production, and, in some cases, death of the cow [6]. The severity and progression of the infection depend on factors such as the pathogen type, the cow's age, immune response, overall health, and stage of lactation [7]. Clinical mastitis can be categorized into subacute, acute, and hyperacute stages based on the degree of inflammation [8]. Hyperacute mastitis involves intense inflammation, a marked drop in milk yield, and altered milk composition, along with systemic signs like fever, depression, chills, and weight loss [9]. Acute mastitis shares similar characteristics but presents milder systemic symptoms, such as lower fever and mild depression. Subacute mastitis, however, shows minimal inflammation and no obvious systemic signs [10]. Severe clinical mastitis can be life-threatening. In sub-clinical mastitis, where there are no apparent udder or milk abnormalities, an increase in somatic cell count (SCC) is associated with reduced milk production [11,12]. These conditions are transmitted through the milker's hands, teat cups, and milking equipment, transferring infectious bacteria from infected to

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uninfected animals [13]. Increased levels of *Corynebacterium bovis*, coagulase-negative staphylococci, *Mycoplasma* spp., *Staphylococcus aureus*, and *Streptococcus agalactiae* considered infected bacteria are associated with mastitis infection in buffaloes [14]. Scientists emphasize that detecting subclinical mastitis (SCM) is crucial for selecting effective strategies for its control, prevention, and treatment [15]. Preventive measures require laboratory diagnostics to isolate and identify the pathogens causing the disease. Mastitis in dairy cattle is caused by various pathogens, including bacteria, mycoplasma, algae, and yeast [16]. While over 200 microorganisms have been identified as mastitis-causing agents, only a few are commonly isolated [17]. Researchers have identified more than 150 species of both Gram-positive and Gram-negative bacteria linked to mastitis [18]. Common Gram-positive pathogens include species of staphylococci and streptococci, while Gram-negative bacteria, such as *E. coli* and *Klebsiella pneumoniae*, are also prevalent. Mastitis is classified into two types based on the mode of infection, typically involving infectious agents [19]. In contrast, the dairy animal's immediate surroundings serve as a reservoir for infection in environmental mastitis. In addition to *Citrobacter*, *E. coli*, *Enterobacter*, *Klebsiella*, and *Streptococcus dysgalactiae*, the most frequent environmental mastitis pathogens include *Bacillus cereus*, *Bacillus licheniformis*, and *Pseudomonas aeruginosa* [20]. Subclinical and chronic mastitis is primarily caused by infectious agents such as *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Streptococcus dysgalactiae*, which colonize the udder skin and spread between animals during milking [21]. In contrast, environmental pathogens like *Escherichia coli*, *Klebsiella pneumoniae*, *Streptococcus uberis*, and *Enterococci* spp., commonly found in soil, water, and bedding, are typically eliminated quickly by the animal's immune system [22]. Increased mastitic pathogens in dairy products reduces milk production in cattle and cheese production from the milk of affected animals. Reduction in the quality and quantity of dairy products hampered dairy economy growth. According to data from different countries, mastitis is one of the dairy cattle diseases with the largest financial burden, especially on small-scale producers in developing countries. In several countries, including India, Bangladesh, Japan, Canada, the United States, Colombia, the Netherlands, the United Kingdom, and Australia, the cost of treating and controlling mastitis is among the highest in the dairy industry [23]. In India, mastitis causes an annual economic loss of about Rs. 575 million and results in a 21 % reduction in milk production [24]. The health of dairy cattle is seriously endangered by antimicrobial resistance (AMR), which commonly arises from the misuse of antibiotics for inhibiting the growth of mastitic bacteria [25]. A notable public health issue stemming from antimicrobial resistance (AMR) in livestock farming is the potential transmission of AMR bacterial infections and drug remnants to humans via the consumption of untreated milk from antibiotic-administered animals. Over billions of years under selective pressure, bacteria have developed a variety of survival mechanisms that allow them to adapt to practically any environmental niche [26]. Biofilm is a preferred maturation stage for more than 90 % of bacteria [27]. Biofilms, intricate microbial communities enmeshed in biopolymer networks, exhibit greater antibiotic resistance than their dispersed planktonic counterparts [28]. The bacteria are defended by their biofilm from external dangers including host defenses and drugs [29]. Understanding the prevalence and characteristics of isolates that form biofilms and are resistant to antibiotics in cases of bovine mastitis may make it possible to utilize antibiofilm drugs as a novel supplemental therapy option and avoid the administration of antibiotics without regard to the patient's specific needs. A critical need exists to reevaluate the conventional antibiotic based treatment approach and seek novel, sustainable therapeutic options. The pursuit of novel approaches in animal healthcare is decreasing the reliance on antibiotics for disease prevention and treatment, thereby alleviating the development of pathogen resistance to these pharmaceuticals. Research has investigated the use of immunization, probiotics, homeopathy, genetic selection, and herbal remedies in place of antibiotics for the treatment and management of mastitis. The primary necessity of the

dairy industry for the production of clean milk is the diagnosis of mastitis [30]. Early detection or therapeutic intervention of mastitis requires a prompt and accurate diagnosis for the prevention and effective management of mastitis. It is imperative to seek potential therapeutic agents as alternatives to antibiotics in the intramammary treatment of bovine mastitis. Therefore, it is wise to consider plants with antimicrobial action against multidrug-resistant (MDR) illnesses as significant resources [31]. Medical herbs have been utilized to treat mastitis because of their antibacterial, anti-inflammatory, immune-stimulatory, and antioxidant qualities [32]. Natural substances or their semi-synthetic equivalents may serve as the basis for new anti-bacterial medications. Compounds present in extracts from medicinal plants, including peptides, unsaturated long-chain aldehydes, essential oils, and alkaloids, hold promise for therapeutic applications in creating new, effective drugs targeting infections that are presently challenging to treat. The present study focuses on the management and advancements in the therapy of mastitis and intramammary infections (IMIs). Additionally, it covers the creation of herbal remedies and natural therapeutics to combat this significant disease affecting the bovine population and dairy herds.

2. Global impact and economic losses

Mastitis is a financially impactful ailment in the dairy sector, affecting both small-scale farmers in developing nations. Additionally, it affects high-yielding cattle herds globally, leading to diverse economic consequences documented in various nations [33]. Economic losses from mastitis encompass direct expenditures related to diagnostic tests, veterinary care, medication, discarded milk, and labor. Moreover, there are additional indirect expenses associated with reduced future milk output, reduced reproductive fitness, premature culling and substitution of afflicted bovines, compromised livestock well-being, and heightened vulnerability to diverse pathogenic conditions [34]. Mastitis affects populations of cattle and buffalo worldwide, posing a severe economic burden for nations like Canada, Germany, India, the Netherlands, the United Kingdom, and the United States of America [35]. Milk production reduction in bovine mastitis varies from 1.0 to 2.5 kg per day during the initial two weeks, depending on factors such as parity and onset time, ultimately resulting in a total decrease of 110–552 kg over the lactation period [36]. Even with the advanced dairy management techniques for cattle and buffalo, mastitis continues to be a serious condition and a major source of concern for dairy farmers and business owners. This problem has a particular impact on India, which produces the most milk on the planet by blending the milk of cattle and buffalo [37]. In India, mastitis results in an annual economic loss of approximately Rs. 575 million, leading to a 21 % diminished milk yield [38]. Moreover, the ingestion of milk contaminated with mastitis-related pathogens has the potential to pose a noteworthy public health threat, as it may facilitate the transmission of antimicrobial-resistant microorganisms. This underscores a substantial concern in the realm of public health. The economic repercussions of mastitis exhibit variability across nations and can further differ within specific regions of a country. This variance is contingent upon factors such as market dynamics, regulatory standards pertaining to milk quality, utilization of diverse diagnostic techniques, and the specific definitions applied for bovine mastitis and intramammary infections (IMI), among other factors. Bovine Mastitis (BM) is predicted to have a negative economic impact on a number of variables, including a 31 % decrease in milk production, a 24 % treatment cost, an 18 % reduction in milk waste, a 4 % rise in labor costs, and a 23 % premature culling of cattle. This collectively results in an estimated annual cost of approximately INR 1.45 to INR 2.4 trillion in the worldwide dairy sector. The annual financial repercussions of Bovine Mastitis (BM) are forecasted to reach approximately INR 147.5 billion in the United States, INR 23.1 billion in Canada, INR 61.7 billion in Colombia, INR 100.7 billion in Australia, around INR 119.3 billion in the European Union, roughly INR 59.7 billion in Japan, about INR 2 million

in Bangladesh, and approximately INR 61.7 billion in India [39]. An isolated instance of clinical mastitis can result in economic losses spanning from approximately INR 9472 to INR 32,776 [40]. In North-West Ethiopia, subclinical mastitis (SCM) affected 33 % of the quarters and 62 % of cows, with a rate of 41.02 % recorded in the Holeta district between August 2011 and May 2012. In Rwanda's peri-urban areas of Kigali, SCM prevalence was 43.1 % at the quarter level and 76.2 % at the cow level, while Rubavu and Nyabihu Districts observed a 50.4 % prevalence at the cow level. In Uganda's urban and peri-urban areas of Kampala, SCM rates ranged from 48.6 % to 86.2 % at the cow level and 19.6 %–55.4 % at the quarter level. Clinical and subclinical mastitis exhibit distinct impacts on milk production, which are also influenced by the specific causative pathogens. The clinical form of mastitis affects around 29 % of cows in Europe and 22 % in North America, while its occurrence is lower in Asia (18 %) and Africa (12 %). Notably, mastitis is responsible for roughly 70 % of the economic losses experienced by dairy farms [8]. The pooled prevalence of clinical mastitis (CM) and subclinical mastitis (SCM) worldwide is 15 % and 42 %, respectively [41]. Across India, the pooled prevalence of CM and SCM was 18 % and 45 %, respectively [39]. Accurate assessment of economic losses and the implementation of customized mastitis control strategies necessitate pathogen-specific data. Therefore, it is imperative to acquire data regarding the prevalence of various mastitis-causing agents within herds. Globally, the prevalence of *Staphylococcus*, *Streptococcus*, and *E. coli* species has been reported to be 28 %, 12 %, and 11 %, respectively, based on 156, 129, and 92 studies. The cost of clinical mastitis (CM) varies by the causative pathogen, with the highest costs associated with Gram-negative bacteria (₹17,640), followed by Gram-positive bacteria (₹11,180), and the lowest costs for other pathogens (₹7970) [42]. Notably, much higher costs were reported for CM, reaching up to ₹27,280, and even ₹37,130 for cases occurring within the first 30 days of lactation. For clinical mastitis, losses primarily stem from culling and replacement costs (48 %), reduced milk production (34 %), unusable milk (11 %), higher labor expenses (3 %), and treatment costs (2 %). In chronic cases, the major costs are reduced milk yield (72 %), culling risk (25 %), and minor costs from discarded milk and veterinary care. Collecting milk samples for bacteriological analysis from both clinical and subclinical mastitis cases would establish a regular source of data regarding the diverse causal agents [43]. Researchers observed an average 25 % reduction in milk yield in cows affected by mastitis, with a 3.5 kg/day loss for *E. coli* infections [44]. The decrease in milk production varied by pathogen, with a 20 % decline for *S. dysgalactiae* and 50 % for *E. coli*. Additionally, milk from cows with clinical mastitis is often deemed unsuitable for processing, resulting in greater economic losses from discarded milk and feeding costs compared to those from reduced milk yields [45]. Subclinical mastitis (SCM) results in significant long-term milk yield losses, as affected cows often produce well below their potential [46]. The impact of SCM extends beyond the period of inflammation, affecting milk production throughout lactation. Although some studies report minimal variation in milk loss due to different SCM pathogens, SCM does alter milk composition [47]. Specifically, reductions in non-fat solids and lactose are noted, with calcium, potassium, inorganic phosphorus, α -lactalbumin, and β -lactoglobulin levels decreasing, while sodium, chloride, pH, albumin, lactate dehydrogenase (LDH) activity, and immunoglobulin levels increase. However, protein and fat content remain largely unchanged, and while SCM increases somatic cell count, it does not significantly affect milk yield [48]. Regrettably, in numerous countries where mastitis is typically managed empirically, without routine sampling, this practice remains restricted. In the context of dairy farm operations, pathogen-related information holds the potential to enhance mastitis control. In scientific investigation, it could unveil new opportunities for identifying pathogen-specific risk factors and the impacts of mastitis, facilitating the development of targeted treatments for implementation on dairy farms.

3. Approaches for the treatment and prophylaxis of mastitis

3.1. Antibiotics

Mastitis infections are the predominant reason for antibiotic usage in the dairy industry. Treatments with antibiotics, which include both natural and synthetic penicillin, is currently the primary approach to managing mastitis. However, escalating expenses, the rise of antibiotic resistance, and declining antibiotic effectiveness in cattle populations have emerged as noteworthy concerns. The excessive and inappropriate use of antibiotics in bovine mastitis poses a significant concern, as it can lead to the development of bacterial resistance and food-chain contamination. The efficacy of currently licensed antibiotics, such as pirlimycin, is constrained in the context of *S. aureus* infection, leading to only about 10–30 % success in curing mastitis [49]. According to a research scientist noted that the available therapies for *S. aureus*-induced mastitis are suboptimal and, consequently, ineffective [50]. Furthermore, the use of antibiotics carries supplementary drawbacks, such as the identification of antibiotic remnants in milk designated for human consumption, eliciting consumer apprehensions [51]. As a result, there exists a critical necessity to investigate innovative antimicrobial interventions, specifically non-antibiotic options, to curtail their utilization in livestock used for food production.

Furthermore, the formation of biofilms is acknowledged as a substantial contributor to antibiotic resistance and recurrent mastitis occurrences. As per research findings, the milk composition exerts an influence on the generation of biofilms by prevalent mastitis pathogens, with a noted inverse relationship between pH levels and biofilm formation [52]. The authors suggest using pH-enhancing drugs to minimize biofilm development and thwart chronic infections given the present reliance on acidic antibiotics for treatment. Given the constraints associated with antibiotics, it is imperative to explore alternative antimicrobial treatments with efficacy against mastitis-causing bacteria urgently. So here this review includes a few potential alternative treatments.

3.2. Potential alternative treatments

3.2.1. Probiotics

Probiotics are gaining increased traction in treating various inflammatory conditions and diseases [53]. Even though many microorganisms have been tested for probiotic qualities, lactic acid bacteria remain the most common kind of probiotics. Lactic acid bacteria exhibit robust immunomodulatory effects, making them effective in mitigating mastitis when utilized as feed supplements, teat dips, or intramammary inoculations [54]. These bacteria inhabit the udder, creating protective biofilms that impede the growth of mastitis-causing pathogens. The health of dairy cows is closely associated with their gut microbiome and the metabolites it produces, particularly lipopolysaccharide (LPS) and short-chain fatty acids [55]. Due to enhanced blood-barrier permeability, elevated LPS from the rumen can enter circulation, reaching the mammary gland and causing inflammation. In contrast, the mammary gland is protected by short-chain fatty acids, a byproduct of rumen fermentation. Incorporating lactic acid bacteria into animal feed is a viable strategy for preventing bovine mastitis. *In vitro* tests revealed the biofilm-forming capacity of several lactic acid bacteria strains [56]. Biofilm formation extent hinged significantly on bacteria strain, surface charge, and medium properties. Out of 165 isolates obtained from teat canal samples, three lactic acid bacterial strains—specifically, *Lactobacillus brevis* 1595, *L. brevis* 1597, and *L. plantarum* 1610, demonstrated robust adherence to bovine mammary epithelial cells. These strains are capable of establishing themselves in the mammary gland and reducing the amounts of pathogens that cause mastitis [6,57]. Through coaggregation and attachment to bovine teat canal epithelial cells, two lactic acid bacterial strains isolated from bovine milk, namely *Lactococcus lactis* subsp. *lactis* CRL 1655 and *Lactobacillus perolens* CRL 1724,

demonstrated inhibitory effects against mastitis-causing pathogens in *in vitro* experiments. This could potentially thwart pathogenic invasion. A negative relationship was identified between *Lactobacillus* levels and mastitis-causing pathogens in mastitic milk. After intramammary administration of *Lactococcus lactis* subsp. *lactis* CRL 1655 and *Lactobacillus perolens* CRL 1724 during the dry-off phase, a notable elevation in immunoglobulin (specifically, IgG isotypes) levels was observed in both blood and milk. This underscores their potent immunomodulatory attributes, fostering both localized and systemic immune responses.

The introduction of lactic acid bacteria into the mammary gland triggered pro-inflammatory responses, leading to an increase in neutrophil presence in milk, both during lactation and the dry-off period [58]. Teat disinfection with an innovative lactobacilli-based probiotic teat dip demonstrated superior efficacy over commercial disinfectants in reducing somatic cell count, attributed to an enhanced microbial environment at the teat. The diminished presence of mastitis-causing bacteria is attributed to the favorable microbial milieu facilitated by environmentally friendly lactic acid bacteria disinfectants, which could serve as substitutes for conventional chemical disinfectants. Probiotics can modify the microbiota at the teat tip, inhibiting the establishment of mastitis-causing bacteria in the teat canal, although the scientific rationale for their use in treating mastitis in dairy calves is not well-established. It is not advised to treat clinical mastitis in lactating dairy cows using probiotics, whether with or without recurrent milk-out. Microbial extracts, in addition to probiotics, show potential in the treatment of mastitis. Antimicrobial substances found in crude extracts from actinomycetes (Caat1-54 and CaatP5-8) inhibited the growth of bacteria linked to both clinical and subclinical mastitis [43,54]. *In vitro* tests using isolates of *S. aureus*, *S. chromogenes*, *S. dysgalactiae*, and *S. uberis* demonstrated the antibacterial activity of these extracts.

3.2.2. Animal-derived substances

Recent research has focused on using bee products to treat bovine mastitis, specifically bee venom containing melittin. The anti-inflammatory impact of melittin was studied by administering it to LPS-induced MAC-T cells [59]. Researchers discovered that bee venom reduced the expression of COX-2 protein induced by LPS. Additionally, it decreased the mRNA expression of inflammatory cytokines TNF- α and IL-6 by inhibiting ERK1/2 phosphorylation and the movement of NF- κ B into the cell nucleus [60]. Chinese propolis, a resin made by honey bees, was investigated for its anti-inflammatory impact on MAC-T cells [61]. Prior exposure of MAC-T cells to Chinese propolis at a concentration of 15 μ g/mL prevented a decline in cell viability. Moreover, it reduced the mRNA levels of pro-inflammatory cytokines like TNF- α and IL-6 when these cells were exposed to different pathogenic elements such as LPS, lipoteichoic acid, TNF- α , heat-inactivated *E. coli*, and *S. aureus*. Additionally, Chinese propolis increased the mRNA expression of antioxidant genes HO-1, TXNRD1, and glutamate-cysteine ligase modifier subunit in mastitis-infected cells, suggesting its antioxidative effects [62]. Conversely, naturally occurring mammalian immunomodulators like lactoferrin were favored as promising non-antibiotic antimicrobial agents for managing and preventing bovine mastitis [63]. Lactoferrin is a versatile glycoprotein that binds iron, present in milk, colostrum, and various exocrine secretions like saliva and tears. As an immunomodulator, it has a crucial role in the innate immune system by enhancing the process of phagocytosis through opsonization of microorganisms [64]. It has been observed to demonstrate antimicrobial properties against *E. coli*, *Pseudomonas aeruginosa*, *Strep. agalactiae*, and *Staph. aureus*. This effect is mainly linked to its capability to chelate iron, which inhibits biofilm formation by sequestering iron.

3.2.3. Nanotechnology in mastitis treatment

Nanotechnology is rapidly advancing and nanoparticle properties find increasing applications in veterinary, pharmaceutical, and medical fields. Due to their distinct chemical and physical characteristics and high surface area-to-volume ratio, nanomaterials provide an option for

controlling pathogens, such as those responsible for dairy bovine mastitis. In order to combat pathogenic microorganisms, silver, gold, and copper nanoparticles are frequently used. The bactericidal and fungicidal attributes of silver nanoparticles stem from their ability to disrupt cell membranes, induce protein denaturation, generate silver ion-enriched microenvironments, impede DNA replication, produce reactive oxygen species, and interfere with respiratory chain proteins and enzymes. Copper nanoparticles exert toxicity against pathogens by inducing reactive oxygen species production, disrupting DNA chains, and causing lipid and protein peroxidation [65]. Conversely, gold nanoparticles alter membrane potential and adenosine triphosphate synthesis activity in pathogenic cells, inflict metabolic inhibition on bacterial strains [66]. The effectiveness of nanoparticle therapy depends on the structural characteristics of the cell walls of pathogenic microorganisms. Copper nanoparticles exhibit higher toxicity towards Gram-positive bacteria with thicker cell walls, while silver and gold nanoparticles prove more efficient against Gram-negative bacteria. As per research findings, *S. aureus* derived from mastitis-afflicted cows was rapidly eliminated when exposed to a 10 μ g/ml solution of silver nanoparticles, with complete destruction occurring in approximately 7 min of incubation. According to research, it has been observed silver nanoparticles to be selectively effective against *E. coli* strains at low concentrations, with lesser toxicity against *S. aureus* [67]. Gram-negative bacteria have been used to illustrate the antibacterial action of silver nanoparticles. Nanosilver displayed potent bactericidal properties, causing damage to *E. coli* cell's structure, disrupting membrane transport, and inducing cell death as revealed by electron microscopy. Research has explored the efficacy of a nanosilver gel against a range of pathogens, including *E. coli*, *S. typhi*, *S. epidermidis*, *S. aureus*, *P. aeruginosa*, and *Candida* yeast-like fungi [68]. Among these, *E. coli* exhibited the lowest minimum inhibitory concentration (MIC) required to kill 50 % of strains at 1.56 μ g/ml, while *S. aureus* had the highest MIC at 6.25 μ g/ml [69]. According to earlier studies, *S. aureus* showed stronger resistance to silver nanoparticles, but *E. coli* and *P. aeruginosa* were more susceptible [70]. The study established that the minimum inhibitory concentration (MIC) of silver nanoparticles was 1 μ g/ml for *P. aeruginosa* and 2 μ g/ml for *S. aureus* isolates obtained from goat milk. Significantly, bacterial strains exposed to nanoparticles showed reduced glutathione expression, decreased superoxide dismutase and catalase activity, but increased glutathione S-transferase expression [71]. It has been demonstrated that with minimum bactericidal concentrations (MBC) ranging from 100 μ g/ml to 5000 μ g/ml, copper oxide nanoparticles have been shown to have the ability to decrease several harmful bacterial populations, including methicillin-resistant *S. aureus* and *E. coli*. Pathogenic bacteria were completely eliminated when copper oxide and silver nanoparticles were used together. Monodisperse copper particles with strong Gram-positive antibacterial activity, similar to nanosilver, were found effective against methicillin-resistant *S. aureus*. They also demonstrated fungicidal effects on *Candida* spp. Additionally, the study examined the antibacterial characteristics of 7-epiclusianone derived from *Rheediabrsiliensis* fruit and its copper metal complex against *Streptococcus* spp. strains responsible for bovine mastitis [72]. The compound exhibited significant prospects for mastitis therapy, with a minimum inhibitory concentration (MIC) of 7.8 μ g/ml and a minimum bactericidal concentration (MBC) ranging from 15.6 to 31.3 μ g/ml against the isolated pathogens. Gold nanoparticles have demonstrated efficacy against fungal pathogens causing mastitis. Nanogold has a lower level of toxicity than other nanoparticles, making its use advantageous. The research findings indicated minimum inhibitory concentrations (MICs) of 2.93 μ g/ml for *E. coli* and 3.92 μ g/ml for *S. aureus* when exposed to 7–34 nm gold nanoparticles [73]. A study assessed the influence of honey and gold nanoparticles on methicillin-resistant *S. aureus* (MRSA) and vancomycin-resistant *S. aureus* (VRSA) strains isolated from mastitis-afflicted cows. According to this research, 30 nm gold nanoparticles at 2.56 μ g/ml concentration inhibited MRSA and VRSA growth, while a nanogold-citrus honey

mixture displayed a 50 % lower MIC. Researchers are presently developing intramammary formulations with metal nanoparticles. The optimistic in vitro outcomes are driving increased interest in subsequent in vivo experiments. Several indications pointing towards the possible introduction of commercial intramammary products containing silver, gold, or copper nanoparticles in the foreseeable future.

3.2.4. Herbal therapy

Contemporary investigations focus on plant-derived compounds for combating bovine mastitis.

The prevailing treatment approach relies on antibiotics, but their efficacy is limited and further complicated by the rise of antibiotic-resistant strains of pathogens, presenting substantial challenges. Escalating apprehensions about antibiotic resistance's impact on public health are compelling dairy industries to curtail antimicrobial drug usage. Therefore, it is crucial to investigate antibiotic therapeutic alternatives, especially those derived from natural sources like plants and animals. Ethno-veterinary medicine, a branch of veterinary science, employs herbal preparations for disease treatment. Plant-derived compounds, renowned for their traditional medicinal uses, are garnering attention from researchers as potential treatments for bovine mastitis. These compounds offer a distinct advantage over antibiotics: they do not promote resistance even with prolonged exposure and possess low toxicity. By altering NF- κ B pathways, several plants have shown to possess antibacterial characteristics and the capacity to reduce inflammation spurred on by pathogens or endotoxins [74]. The assessment of antimicrobial activity involves techniques such as paper disc assay and MIC assay. The antibacterial properties of leaf extracts from *Nyctanthes arbor-tristis* Linn, also known as jasmine or parijatha, were evaluated in a study against pathogens that cause mastitis [75]. The study encompassed in vitro assessments of aqueous and methanolic extracts derived from *Nyctanthes arbor-tristis* leaves, both in isolation and in combination, with a reference to the antibiotic ciprofloxacin employed as a positive control [76]. The results showed that *S. aureus* and *Escherichia coli* were most effectively inhibited by the aqueous extract solely whereas the mixture of aqueous and methanolic extracts had the strongest inhibitory effect on *Streptococcus agalactiae*. *Nyctanthes arbor-tristis* leaf extracts and their combination demonstrated notable antimicrobial activity, with minimum inhibitory concentration (MIC) values against *Streptococcus agalactiae* and *Staphylococcus aureus* were 50 μ g/mL and 25 μ g/mL, respectively. In the case of *E. coli*, both the methanolic and mixed extracts exhibited equivalent minimum inhibitory concentration (MIC) values of 50 μ g/mL, whereas the aqueous extract demonstrated a reduced MIC of 12.5 μ g/mL [77]. Furthermore, the aqueous extract demonstrated bactericidal efficacy against *Streptococcus agalactiae*, with a minimum bactericidal concentration (MBC) of 100 μ g/mL. In contrast, the methanolic and mixed extracts showed the highest MBC values of 50 μ g/mL against *Escherichia coli* and *Streptococcus agalactiae*. A research study aimed to assess the antibacterial properties of aqueous and alcoholic extracts derived from a variety of medicinal plants, including *Dactyloctenium indicum*, *Argemone Mexicana*, *Calotropis gigantea*, *Trigonella foenum-graecum*, *Acacia leucophloea*, *Acacia nilotica*, *Achyranthes aspera*, *Acalypha indica*, *Coccinia grandis*, *Asteracantha longifolia*, *Abutilon indicum*, *Cenchrus ciliaris*, *Brachiaria* sp., *Trichodesma indicum*, *Melinis repens*, *Barlerialupulina*, *Chloris inlata*, *Medicago sativa*, *Spermacoce hispida*, and *Rhynchosiacapitata*. The research investigated the antibacterial efficacy of two extracts against *K. pneumoniae*, *E. coli*, *S. aureus*, and *S. agalactiae*. The observed zones of inhibition against these microorganisms varied between 8.0 and 25.0 mm. Furthermore, the extracts demonstrated minimum inhibitory concentration (MIC) values spanning from 0.125 to 2.00 mg/ml. The research discovered that silver nanoparticles (AgNPs) synthesized using *Moringa oleifera* and *Tamarindus indica* were similarly effective against *S. aureus* [78]. Furthermore, the antibacterial activity of these AgNPs closely resembled that of the antibiotics examined in the study. According to a study, *B. aegyptica*, *T. indica*, and *C. toka* condensed tannin extracts have substantial

antibacterial activities. Specifically, the condensed tannins derived from the foliage of *B. aegyptica* and *T. indica* exhibited antimicrobial activity equivalent to or on par with commercial penicillin. Further investigation is required to elucidate which specific monomers of these condensed tannins, such as catechin, epicatechin, and their galloyl derivatives, contribute to the antibacterial actions against mastitis-causing microorganisms. The antibacterial properties of aqueous extract from *Balanites aegyptiaca* fruits (AE-BAF) and ethanolic extract from *Curcuma longa* powder (EE-CLP) were examined at various concentrations against bacteria isolated from collected milk samples [79]. The identified bacteria included *Staphylococcus aureus*, *Escherichia coli*, *Streptococcus agalactiae*, *Salmonella* spp., *Bacillus subtilis*, *Klebsiella pneumoniae*, and *Pseudomonas aeruginosa*. A research study examined the possible application of *Ocimum sanctum* (holy basil) leaf juice as an adjunct therapy for experimentally induced chronic staphylococcal mastitis. The investigation disclosed notable bio-enhancing and antioxidant characteristics associated with the leaf extract [80]. These attributes could be effectively harnessed through combination with antibiotics for therapeutic purposes.

Natural compounds offer advantages such as their origin from natural sources, minimal side effects and reduced resistance development with prolonged exposure compared to conventional drugs. The ethyl acetate extract of *Terminalia chebula* is one example of a plant-derived chemical that has been shown in several trials to be efficacious [81]. The extract's efficacy, comparable to amoxicillin (100 μ g/mL) at a concentration of 500 μ g/mL, demonstrated a broad-spectrum antibacterial effect against all bacterial strains isolated from subclinical mastitis in cows. According to Ranjith et al. (2018), a methanolic extract made from the herbal mixture that contained equal amounts of *Curcuma longa* rhizomes and *Diplocisiaglaucescens* leaf had both analgesic and anti-inflammatory properties. This herbal extract was discovered to have analgesic activities comparable to ibuprofen and indomethacin. Herbal therapy encompasses various administration routes, including topical application, oral ingestion, and intramammary delivery, with each route chosen based on the specific formulation and application method. In vitro experiments were conducted to assess the impact of *Oxytropis glabra* decoction on the biofilm formation of *S. epidermidis* [82]. These investigations revealed promising inhibitory mechanisms that merit further investigation for the development of innovative drugs targeting infections associated with biofilms. The traditional Chinese medicinal *Sanguisorba officinalis* L. also showed the capacity to suppress biofilm development in methicillin-resistant *Staphylococcus aureus* (MRSA) strains, underlining its promising antibacterial characteristics. Furthermore, these plant-derived compounds were observed to have a detrimental impact on the formation of bacterial biofilms. Biofilms are complex bacterial assemblies featuring polysaccharides or proteins, conferring several benefits to bacteria. These advantages encompass facilitating attachment and growth, granting resilience against elements such as phagocytosis, antimicrobial substances, and disinfectants. These benefits primarily result from restricted diffusion within the biofilm matrix and modifications in cellular metabolism. Moringa extract has demonstrated the ability to reduce inflammatory reactions and enhance antioxidant defenses in the epithelial cells of the bovine udder [83]. It significantly decreased the expression of proinflammatory cytokines (TNF- α , IL-1 β , and IL-6) and cyclooxygenase-2, along with regulating NF- κ B. Moreover, it elevated the expression of heme-oxygenase-1, NAD (P)H, and quinone oxidoreductase-1. The Moringa plant extract also stimulated the synthesis of casein proteins. Propolis exhibited notable antimicrobial effectiveness against mastitis-related pathogens. Propolis was most effective against Gram-positive bacteria, except for *C. bovis* and *E. faecalis*, while Gram-negative bacteria, including *Pasteurella* spp. and certain *E. coli* strains, displayed lower susceptibility [84]. In a table summarize the application of various botanical species for bovine mastitis treatment systematically cataloged by their botanical name, common name, plant family, utilized plant components, bioactive extracts, their respective therapeutic actions, and the primary targeted

bacteria with restricted or inhibited growth (Table 1).

4. Management of mastitis

Effective mastitis management involves a comprehensive approach to mitigating intramammary infections (IMIs) in dairy cattle. Rather than solely focusing on IMI treatment, a proactive strategy aims to prevent mastitis occurrence. Essential to this approach are robust management practices, spanning cattle selection to nutritional considerations. This holistic approach is pivotal in achieving successful mastitis prevention.

4.1. Dry cow therapy

The use of antibiotics in dry cow therapy is recommended as an effective strategy for controlling intra-mammary infections and preventing the progression of mastitis [114]. The dry period is a crucial phase during which dairy cows are particularly vulnerable to acquiring new intra-mammary infections, which can originate from both contagious and environmental sources. Notably, approximately 61 % of novel infections arise during this interval. The administration of antibiotics during the dry period offers advantages by obviating the necessity to withhold milk from treated quarters. Antibiotics are given at the conclusion of lactation to maintain sustained udder concentrations over a period of 20–70 days for the eradication of pathogenic bacteria [115]. The therapeutic antibiotic's heightened penetration ability results from extended contact time, effectively treating intra-mammary infections unless pathogenic resistance emerges. Dry cow therapy has exhibited near-complete efficacy in resolving mastitis induced by *S. agalactiae* [116]. In contrast, its efficacy against *S. aureus*-induced mastitis is comparatively lower than that against streptococcal mastitis. The efficacy of dry cow therapy in managing *S. aureus* mastitis is approximately 50 %, but vaccinating against this pathogen during the dry phase has the potential to enhance antibiotic effectiveness. A two-week duration of dry cow treatment has demonstrated marked reductions in clinical mastitis instances attributed to *S. dysgalactiae* and *S. uberis* infections.

4.2. Genetic selection

Distinct genetic variations are evident among diverse dairy breeds; Holstein, for instance, typically exhibits elevated somatic cell counts (SCC) compared to central European dairy breeds [117]. Furthermore, intrabreed variations are also discernible, influencing productivity and disease resilience, including mastitis. Characterizing these genetic variations aids in identifying cattle possessing genetic traits resistant to mastitis, which can be used for selective breeding. Certain genomic mutations/genotypes, particularly those related to lactoferrin (encoded by BTA22), have gained recognition for their possible role in mastitis resistance, given their bacteriostatic properties in mammary glands [115]. In a comparable manner, BTA5 contains genetic information for lysozymes, which play a role in breaking down bacterial cell walls and facilitating the clearance of udder infections. Additionally, the MHC and BoLA (bovine major histocompatibility complex), in conjunction with BTA23, influence mastitis resistance by regulating adaptive immune responses against udder pathogens [118]. However, the precise roles of these factors remain elusive in many instances. Genetic predilections influencing teat morphology and dimensions underscore herd genetic diversity. The configuration of teat structure has been demonstrated to affect the susceptibility of cows to mastitis. As a result, crossbred cows with pointed or round teat ends are more prone to post-milking tissue alterations when exposed to mechanical milking, in contrast to cows with flat-ended teats.

4.3. Nutrition

The correlation between animal nutrition and mammary tissue's

resistance to infection is attributed to nutrients antioxidant properties, enhancing immune defense against infections. The health of the udder is influenced by important trace elements including copper, zinc, and selenium as well as vitamins like E and A/b-carotene. The administration of trace minerals such as zinc, manganese, selenium, and copper has shown to reduce the incidence of chronic clinical mastitis in cows exhibiting elevated somatic cell counts (SCC) [119]. An energy deficit in cows elevates the probability of ketosis, resulting in a twofold increase in the susceptibility to clinical mastitis. Copper supplementation in cows during intramammary *E. coli* infection decreased bacterial load and somatic cell counts (SCC), thereby mitigating the severity of the *E. coli* infection [120]. Selenium demonstrates anti-inflammatory effects through the modulation of inflammatory mediators. Dietary selenium supplementation in mice inhibited NALP3 inflammasome activation and NF- κ B/MAPK pathways, reducing mastitis induced by *S. aureus* [121]. A combination of selenium and vitamin E elevates neutrophil α -tocopherol concentrations, improving mastitis resistance. Vitamins A, D3, E, and H aid subclinical mastitis recovery by increasing host defense gene expression. Vitamin D activates bovine monocyte innate immunity and normalizes oxidant-antioxidant balance. Vitamin E and selenium boost phagocytic function, thereby decreasing the susceptibility to mastitis. Adequate vitamin E intake through fresh grass and silage is important. Antioxidant supplementation like vitamins A and E benefits mammary gland health. Dietary supplements, such as fatty acids and cholecalciferol, modulate gene expression and the internalization of bacteria. Zeolite clinoptilolite supplementation affects milk composition and udder health [122]. Targeted nutrition strategies can limit antimicrobial use, avoiding drug-related issues and promoting positive host responses in disease management. Further research is needed to delineate the connection between nutrition levels and mastitis resistance.

4.4. Lactation therapy

Lactation therapy, involving antibiotic treatment for mastitis during lactation, is employed to alleviate clinical mastitis symptoms and restore normal milk production in cows. This method proves efficacious in decreasing milk somatic cell counts (SCC) while preserving milk quality. Nonetheless, using this therapy for subclinical mastitis is not economically viable due to limited efficacy. The effectiveness of lactation therapy is influenced by factors such as milk somatic cell count (SCC) at the time of treatment, lactation stage, the cow's immune status, age, pathogen strain, and infection type [123]. Newly acquired intramammary infections display a more favorable response to antibiotics compared to chronic infections, and younger cows demonstrate superior treatment outcomes when contrasted with older counterparts. Spontaneous healing is rare in cases of clinical and subclinical mastitis caused by *S. aureus* with lactation therapy, and chronic clinical mastitis cases attributed to this pathogen generally do not significantly respond to such treatment [124]. Extended antibiotic treatment at therapeutic levels may enhance cure rates for clinical *S. aureus* mastitis. A drawback of this therapy is milk loss due to antibiotic residues.

4.5. Teat sealer

Advancements in the formulation of teat sealants, whether applied internally or externally, during the dry period hold promise for mastitis management and the reduction of its consequences. External sealants like DryFlex and Delaval exhibit potential in diminishing new mammary gland infections during dry periods [125]. However, the limited persistence of external sealers is a notable drawback. In real-world situations, bismuth subnitrate has been shown to significantly reduce the frequency of new infections by 10 times when used as an internal teat sealer [126]. When combined with prolonged-action antibiotics in dry periods, internal teat sealers lead to a 30 % reduction in new intramammary infections and a 33 % decrease in clinical mastitis cases. The concurrent usage of bismuth subnitrate and cloxacillin in dry cow

Table 1

Showing different plants employed in bovine mastitis treatments are systematically listed by their botanical names, accompanied by common names, plant families, plant part used, plant extracts and their activities and their effects on targeted bacteria.

S. No	Medicinal plants (Botanical Name)	Common Names	Plant Family	PlantPart used	Plant extracts and their activities	Bacteria mainly restricted or inhibited	Reference
1	<i>A. Mexicana</i>	Satyanashi	Papaveraceae	Leaves	200 mg/ml of methanolic extract showed a zone of inhibition and MIC conc. against <i>S. aureus</i> (15 mm, 0.25 mg/ml), <i>E. coli</i> (10 mm), <i>S. agalactiae</i> (12 mm, 5 mg/ml).	<i>S. aureus</i> , <i>E. coli</i> , <i>S. agalactiae</i>	[85]
2	<i>A. Sativum</i>	Garlic	Amaryllidaceae	Freeze dried garlic	0.8–40 mg/ml methanolic extract showed a zone of inhibition and MIC conc. against <i>S. aureus</i> (15 mm, 0.25 mg/ml), <i>E. coli</i> (10 mm), <i>S. agalactiae</i> (12 mm, 5 mg/ml).	<i>S. aureus</i> , <i>E. coli</i> , <i>S. agalactiae</i>	[86]
3	<i>Azadirachata indica</i>	Neem	Meliaceae	Leaves	MIC of methanolic extract of leaves was showed against 31.25 µg/ml for both <i>S. agalactiae</i> and <i>E. Coli</i> and 15.62 µg/ml for <i>S. aureus</i> .	<i>S. aureus</i> , <i>M. pyogenes</i> , <i>S. agalactiae</i> and <i>E. coli</i>	[87]
4	<i>Alpinia zerumbet</i>	Shell ginger	Zingiberaceae	Leaves	MIC of methanolic extract of leaves was showed against 16 mg/ml for both <i>S. agalactiae</i> and <i>E. Coli</i> and 8 mg/ml for <i>S. aureus</i> .	<i>S. aureus</i> , <i>E. coli</i> ,	[88]
5	<i>Bidens pilosa</i>	Cobbler pegs	Asteraceae	Shoot	Extracts showed antibacterial activity against <i>S. aureus</i> (MIC = 0.3 mg/mL).	<i>S. aureus</i> , <i>E. coli</i> ,	[89]
6	<i>Brachiaria</i> sp.	Bread grass	Poaceae	Leaves	Methanolic extract showed MIC value 0.125–2.00 mg/ml and zone of inhibition is 8–25 mm against <i>E.coli</i> .	<i>S. aureus</i> , <i>E. coli</i> , <i>K. pneumoniae</i> , <i>Streptococcus agalactiae</i>	[90]
7	<i>C. ciliaris</i>	Buffel grass	Poaceae	Whole plant	200 mg/ml of methanolic extract showed a zone of inhibition and MIC conc. against <i>S. aureus</i> (24 mm, 0.125 mg/ml), <i>E. coli</i> (16 mm, .250 mg/ml), and <i>S. agalactiae</i> (5 mm).	<i>S. aureus</i> , <i>E. coli</i> , <i>S. agalactiae</i>	[91]
8	<i>C. grandis</i>	Kowai fruit	Cucurbitaceae	Leaves	200 mg/ml of methanolic extract showed a zone of inhibition and MIC conc. against <i>S. aureus</i> (25 mm, 0.125 mg/ml), <i>E. coli</i> (17 mm, 0.5 mg/ml), <i>S. agalactiae</i> (9 mm, 2 mg/ml).	<i>S. aureus</i> , <i>E. coli</i> , <i>S. agalactiae</i>	[92]
9	<i>C. sancta</i>	Holy hawksbeard	Erebidae	Leaves and flowers	The methanol extract of <i>C. sancta</i> showed MIC (minimum inhibition concentration) value for <i>S. aureus</i> was determined as 3250 µg/mL.	<i>S. aureus</i>	[93]
10	<i>Chrysanthemum indicum</i>	Indian Chrysanthemum	Asteraceae	Flower	Aqueous extract showed anti- bacterial activity and provides MIC (3.75 %) and MBC (1.87 %) value for <i>E. coli</i> .	<i>Clostridium perfringens</i> , <i>Listeria monocytogenes</i> , and <i>Escherichia coli</i> .	[94]
11	<i>Curcuma longa</i>	Turmeric	Zingiberaceae	Rhizome	Ethanol extract showed MIC was 11.5 µg/ml for <i>S. aureus</i> .	<i>Staphylococcus aureus</i> , <i>E. coli</i>	[95]
12	<i>Cymbopogon citratus</i>	Lemon grass	Poaceae	Leaves	Oil extract showed zone of inhibition against <i>S. aureus</i> is 14.33 mm (MIC 0.03 %) and <i>E.coli</i> 8.33 mm (MIC 0.06 %).	<i>S. aureus</i> , <i>Staphylococcus</i> positive coagulase, <i>Staphylococcus</i> negative coagulase, <i>Streptococcus uberis</i> , <i>P. aeruginosa</i> and <i>E. coli</i>	[96]
13	<i>Emblica officinalis</i>	Indian gooseberry	Phyllanthaceae	Fruit	<i>Staphylococcus aureus</i> and <i>Escherichia coli</i> were both susceptible to the antibacterial effects of aqueous extract. It displayed a zone of inhibition of 19.7 mm and 18.81 mm, and the MIC and MBC were found to be 5.62 % and 11.25 %, respectively, whereas the observed values against <i>E. coli</i> were found to be 22.5 % and 45 %.	<i>S. aureus</i> , <i>E. coli</i>	[97]
14	<i>Hibiscus rosa sinensis</i>	China Rose	Malvaceae	Flower	The MIC value for <i>S. aureus</i> in ethanol extract was found to be 3.75 %, as well as MBC values of 1.87 % mg/ml for <i>S. aureus</i> , <i>Bacillus cereus</i> , and 3.75 % for <i>E. coli</i> .	<i>S. aureus</i> , <i>B. cereus</i> , <i>P. aeruginosa</i>	[98]
15	<i>J. montana</i>	Iron flowers	Campanulaceae	Flower	The MIC value for <i>S. aureus</i> in ethanol extract was found to be 0.03–0.14 % as well as MBC values of 1.87 % mg/ml for <i>S. aureus</i> , <i>Bacillus cereus</i> , and 3.75 % for <i>E. coli</i> .	<i>S. aureus</i> , <i>E. coli</i> , <i>Streptococcus</i> spp., <i>Streptococcus agalactiae</i> , <i>Pseudomonas</i> spp. and non- <i>aureus</i> staphylococci	[99]
16	<i>Levandula stoeches</i>	Spanish lavender	Lamiaceae	Leaves	Methanolic; MIC conc. for <i>L. stoeches</i> is 3250 µg/ml and the zone of inhibition is around 10 mm for <i>S. aureus</i> .	<i>S. aureus</i> , <i>E.coli</i>	[100]
17	<i>M. sativa</i>	Alfalfa	Poaceae	Whole plant	200 mg/ml of methanolic extract showed a zone of inhibition and MIC conc. against <i>S. aureus</i> (13 mm, 0.5 mg/ml), <i>E. coli</i> (10 mm), <i>S. agalactiae</i> (10 mm).	<i>S. aureus</i> , <i>E. coli</i> , <i>S. agalactiae</i>	[101]
18	<i>Malva parviflora</i>	Small-flowered mallow	Malvaceae	Leaves	MIC value and zone of inhibition were determined for <i>S. aureus</i> (160 µg/ml,	<i>S. aureus</i> , <i>E. coli</i> , <i>P. aeruginosa</i>	[101]

(continued on next page)

Table 1 (continued)

S. No	Medicinal plants (Botanical Name)	Common Names	Plant Family	PlantPart used	Plant extracts and their activities	Bacteria mainly restricted or inhibited	Reference
19	<i>Mentha Longifolia</i>	Horse mint	Lamiaceae	Leaves and roots	approx 13 mm), <i>E. coli</i> (80ug/ml, approx 8 mm), and <i>P. aeruginosa</i> (40 µg/ml, approx 7 mm). MIC value and zone of inhibition determined for <i>S. aureus</i> (40 µg/ml, approx 8 mm), <i>E. coli</i> (40ug/ml, approx 11 mm), and <i>P. aeruginosa</i> (40 µg/ml, approx 10 mm).	<i>S. aureus</i> , <i>E. coli</i> , <i>P. aeruginosa</i>	[102]
20	<i>Murraya koenigii</i>	Curry leaf	Rutaceae	Leaves	Alcoholic extract had a minimum inhibitory concentration (MIC) of 31.25 mg/ml and 125 mg/ml, respectively, and a zone of inhibition against <i>S. aureus</i> and <i>E. coli</i> of 14.13 mm and 16.00 mm, respectively.	<i>S. aureus</i> and <i>E. coli</i>	[103]
21.	<i>Minthostachys verticillata</i>	Peperina	Lamiaceae	Leaves	Methanolic extract showed MICvalue for <i>Streptococcus agalactiae</i> is 50 µg/mL and for <i>S. aureus</i> is 25 µg/mL.	<i>Streptococcus uberis</i> and <i>S. aureus</i>	[104]
22.	<i>Nepata cataria</i>	Catnip	Lamiaceae	Leaves	Ethanollic extract creates Zone of inhibition for <i>E. coli</i> is 10 mm, <i>S. aureus</i> is 8 mm at MIC conc. 0.75 mg/ml.	<i>S. aureus</i> , <i>E. coli</i> , <i>S. agalcticae</i>	[105]
23.	<i>Nyctanthes arbortristis</i> Linn.	Night blooming Jasmine	Oleaceae	Leaves	Methanolic extract showed MICvalue for <i>Streptococcus agalactiae</i> is 50 µg/mL and for <i>S. aureus</i> is 25 µg/mL.	<i>S. aureus</i> , <i>E. coli</i> , <i>Streptococcus agalactiae</i>	[106]
24.	<i>Ocimum sanctum</i>	Holi basil	Lamiaceae	Leaves	Aqueous/Alcoholic extract showed anti-microbial activities. It showed inhibitory zones and MIC value for <i>E. coli</i> is 17.38 ± 0.92mm125 mg/ml and for <i>Streptococcus</i> spp. is 10.88 ± 0.58 mm and 62.5 mg/ml.	<i>Streptococcus</i> spp. And <i>E. coli</i>	[107]
25.	<i>Pulicaria undulata</i>	Desert golden daisy	Asteraceae	Flowers	MIC value and zone of inhibition determined as for <i>S. aureus</i> (160 µg/ml, approx 11 mm), <i>E. coli</i> (80ug/ml, approx 8 mm), <i>P. aeruginosa</i> (40 µg/ml, approx 9 mm).	<i>S. aureus</i> , <i>E. coli</i> , <i>P. aeruginosa</i>	[108]
26.	<i>P. agranatium</i>	Pomegranate	Punicaceae	Leaves, roots	Methanolic extract of <i>S. campanulate</i> showed inhibitory activities and showed MIC value (0.125 µg/ml) against <i>S. aureus</i> .	<i>S. aureus</i> , <i>S. saprophyticus</i> and non-aureus staphylococci	[109]
27.	<i>S. hispida</i>	Shaggy buttonweed	Rubiaceae	Whole plant	200 mg/ml of methanolic extract showed zone of inhibition and MIC conc. against <i>S. aureus</i> (19 mm, 0.50 mg/ml), <i>E. coli</i> (18 mm), <i>S. agalactiae</i> (5 mm).	<i>S. aureus</i> , <i>E. coli</i> , <i>S. agalactiae</i>	[110]
28.	<i>Spathodea campanulate</i>	African tulip tree	Bignoniaceae	Bark	Methanolic extract of <i>S. campanulate</i> showed inhibitory activities and showed MIC value (0.125 µg/ml) against <i>S. aureus</i> .	<i>S. aureus</i>	[111]
29.	<i>T. indicum</i>	Indian borage	Boraginaceae	Whole plant	200 mg/ml of methanolic extract showed a zone of inhibition and MIC conc. against <i>S. aureus</i> (15 mm, 0.250 mg/ml), <i>E. coli</i> (15 mm, 0.25 mg/ml), <i>S. agalactiae</i> (16 mm, 0.2 mg/ml)	<i>S. aureus</i> , <i>E. coli</i> , <i>S. agalactiae</i>	[112]
30.	<i>Tridax procumbens</i> Linn.	Coat buttons	Asteraceae	Leaves	The methanolic extract showed anti-bacterial growth (<i>Staphylococcus aureus</i> MTCC 87) at a concentration of 0.8 mg/ml.	<i>Staphylococcus aureus</i>	[113]

therapy has proven to be effective in addressing both clinical and sub-clinical mastitis. The effectiveness of OrbeSeal, an internal sealing product, in decreasing new mammary gland infections in lactating animals is well-documented through multiple research studies.

4.6. Acoustic pulse therapy

A portable device is used in Acoustic Pulse Therapy (APT), sometimes referred to as shockwave therapy, to generate pulsating pressure waves [127]. These waves deeply penetrate tissues, capable of disrupting chronic wound scar tissue, promoting revascularization. A study focusing on subclinical mastitis demonstrated that APT led to a notable proportion (70.5 %) of mastitis-affected cattle restoring normal milk production, surpassing the control group (18.4 %). APT-treated cows exhibited significantly higher occurrences of milk with a low log SCC (somatic cell count) below 5.6 cells/mL compared to controls [38]. Notably, APT effectively treated cows infected with different bacterial strains, albeit with varied cure rates based on the bacterial type. In cases of clinical mastitis, APT achieved a considerable cure rate (76.9 %) in

affected cows (n = 13), while the gold standard antibiotic therapy only achieved a 18.7 % cure rate. Despite the encouraging results, further comprehensive investigations, encompassing diverse infectious agents, are warranted for therapy standardization. Moreover, exploring the synergistic potential of combining APT with Class IV laser and interferential therapy may enhance healing prospects.

4.7. Utilization of smartphone applications and messaging services

The substantial progress in communication technology evolution has brought notable advancements to the dairy industry in multiple sectors. Communication technology has been utilized for a number of objectives, such as providing breeders with information and enabling the dissemination of educational guidelines. Mastitis control initiatives were instituted, furnishing daily updates to dairy breeders in Kenya and Indonesia [128]. Kenyan farmers receiving investigatory insights through one-way SMS received daily information, although limited mobile memory prompted the deletion of older messages; relevant messages were retained. It has been utilized for livestock health monitoring,

particularly in mastitis diagnosis and estrus detection. It finds utility in broader disease diagnosis. Communication technology is employed through nano-biosensors interfaced with smartphone apps for swift identification of milk sample pathogens. Sensors integrated with apps within automated milking systems offer real-time updates on udder health and milk composition. The digitalization of milking machine controls has contributed to reduced clinical mastitis prevalence due to enhanced machine control. Alternatively, communication technology serves to accumulate epidemiological data concerning regional dairy herds for epidemiological investigation.

5. Conclusion and prospects for future research

Mastitis is a major economic issue in the dairy industry, affecting both small farms and large-scale dairy herds globally. Subclinical mastitis, which is harder to detect, has a significant impact on milk quality and production. Although antibiotics are commonly used for treatment, their prolonged use has led to increased drug resistance and limited treatment efficacy. Factors such as structural changes in udder tissue, mastitis-causing bacteria, and poor livestock management contribute to this limitation. Consequently, researchers are focused on developing alternative treatments and more effective prevention strategies to combat mastitis. Future research should prioritize innovative treatments such as probiotics, nanotechnology, and herbal remedies to address the current challenges. This review highlights the global impact and advancements in managing mastitis and intramammary infections (IMIs), emphasizing the development of herbal and natural therapies. Nanotechnology plays a pivotal role in advancing treatment strategies for bovine mastitis by leveraging the unique antimicrobial properties of silver, gold, and copper nanoparticles. These nanoparticles effectively target mastitis-causing pathogens, exhibiting bactericidal and fungicidal activity against both Gram-positive and Gram-negative bacteria, with potential applications in intramammary formulations. Herbal-derived nanoparticles enhance the bioavailability and antimicrobial efficiency of natural compounds, offering a potent alternative to conventional treatments. The promising in vitro results highlight the potential for future in vivo studies and the development of commercial nanoparticle-based therapies for mastitis control. Notably, plant-derived compounds hold promise as novel antimicrobial agents, demonstrating efficacy against drug-resistant microorganisms both in vitro and, in certain instances, in vivo, with a notable emphasis on their safety for animals, plants, and the environment. The primary benefit of these therapeutic approaches is their potential to address the growing issue of antimicrobial resistance. Some plants have demonstrated in vivo efficacy, such as *Ocimum sanctum* (L). Intramammary infusion of its aqueous extract significantly increased neutrophils, lymphocytes, and lysosomal enzyme content in milk polymorphonuclear cells, thereby enhancing mammary immunity. Oral administration of leaf powder at a dose of 600 mg/kg body weight for 7 days led to a significant reduction in somatic cell count and ceruloplasmin levels, reducing udder inflammation and exhibiting immunomodulatory effects in cows with subclinical mastitis. Intramammary infusions of the hydromethanolic extract of *Azadirachta indica* demonstrated anti-inflammatory, antibacterial, and immunomodulatory properties. Abboud et al. (2015) found that essential oils from *Thymus vulgaris* and *Lavandula angustifolia* exhibited strong antibacterial activity against *Staphylococcus* and *Streptococcus* species in an in vivo study. Intramammary application of a 10 % methanol solution of these oils, individually or combined, significantly reduced bacterial counts in milk samples after four days of treatment. The most effective results were achieved with external application of thyme essential oil in vaseline, leading to a 100 % recovery rate. Herbal treatments, which primarily utilize plant extracts and essential oils, have shown promising results in controlling mastitis pathogens. Their natural origin, reduced side effects, and lower likelihood of resistance development compared to conventional drugs offer significant advantages. Diagnostic methods and treatment strategies should be concurrently advanced to facilitate

prompt, precise, on-farm diagnosis, which can be integrated with targeted therapies against identified microorganisms. This integration is pivotal for enhancing the effectiveness of mastitis control and treatment. Prior to implementing alternative therapies on a commercial scale, it is advisable to conduct field studies for the purpose of validating the treatment outcomes.

CRediT authorship contribution statement

Sonal Sindhu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Tarun Saini:** Writing – review & editing, Formal analysis. **Harsh Kumar Rawat:** Visualization, Software. **Manjeet Chahar:** Methodology, Data curation. **Ankita Grover:** Visualization, Formal analysis. **Sayed Ahmad:** Visualization, Validation, Supervision, Formal analysis, Data curation, Conceptualization. **Hari Mohan:** Visualization, Validation, Supervision, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

This review investigates Mastitis, an inflammatory condition in mammary glands caused by infectious agents, posing a critical challenge in the dairy sector, impacting animal welfare and causing substantial economic losses. The exploration delves into the efficacy of botanical extracts and essential oils with antimicrobial properties as novel approaches for managing mastitis pathogens. It highlights the limitations of current antibiotic treatments and advocates for the development of novel therapeutic alternatives. By emphasizing the progression of herbal remedies and natural therapeutics, this review addresses the significant concern of mastitis in bovine populations and dairy herds, offering insights into effective management strategies.

We anticipate that this review will captivate the readership of your journal due to its valuable insights and contributions to the field. We are optimistic that our research will enrich the expanding corpus of literature concerning alternative treatments, particularly focusing on natural therapeutics, as substitutes for antibiotics, alongside efficacious management tactics.

There is no conflict of interests between the authors. There is no animal as well as human study used in the study.

References

- [1] E. Klasky, Milking welfare: agricultural technology and subject formation on the industrial dairy farm. <https://dx.doi.org/10.7302/4328>, 2022.
- [2] A.S. Neculai-Valeanu, A.M. Arton, Udder health monitoring for prevention of bovine mastitis and improvement of milk quality, *Bioengineering* 9 (11) (2022) 608, <https://doi.org/10.3390/bioengineering9110608>.
- [3] M. Samad, Review on mastitis in dairy lactating animals and their public health importance: the 56 years Bangladesh perspective, *J. Vet. Med. OH Res* 4 (2) (2022) 33–114, [https://doi.org/10.36111/jvmohr.2022.4\(2\).0033](https://doi.org/10.36111/jvmohr.2022.4(2).0033), 10.36111/jvmohr.2022.4(2).0033.
- [4] A. Argaw, Review on epidemiology of clinical and subclinical mastitis on dairy cows, *Food Sci. Qual. Manag.* 52 (6) (2016) 56–65, Giesecke, W. H., den Heever, V., & Wepener, L. (1974).
- [5] G.V. Fabuel, Predicting Metritis Events in Dairy Cattle Using Machine Learning Classifiers on Multiple Data Streams under Nowcasting and Forecasting Frameworks, University of California, Davis, 2021.
- [6] X. Zhao, P. Lacasse, Mammary tissue damage during bovine mastitis: causes and control, *Journal of animal science* 86 (suppl.13) (2008) 57–65, <https://doi.org/10.2527/jas.2007-0302>.
- [7] V. Bronzo, V. Lopreiato, F. Riva, M. Amadori, G. Curone, M.F. Addis, B. Castiglioni, The role of innate immune response and microbiome in resilience of dairy cattle to disease: the mastitis model, *Animals* 10 (8) (2020) 1397, <https://doi.org/10.3390/ani10081397>.
- [8] D. Mota-Rojas, D. Wang, C.G. Titto, J. Gómez-Prado, V. Carvajal-De la Fuente, M. Ghezzi, J. Martínez-Burnes, Pathophysiology of fever and application of infrared thermography (IRT) in the detection of sick domestic animals: recent advances, *Animals* 11 (8) (2021) 2316, <https://doi.org/10.3390/ani11082316>.
- [9] W.H. Giesecke, V. den Heever, L. Wepener, The diagnosis of bovine mastitis with particular reference to subclinical mastitis: a critical review of relevant literature. <http://hdl.handle.net/2263/53798>, 1974.
- [10] N. Saleh, T. Allam, A. Omran, A. Mohamed, A review of the subclinical mastitis in cattle with special reference to the new approaches of its diagnosis and control,

- Journal of Current Veterinary Research 4 (1) (2022) 33–46, <https://doi.org/10.21608/jcvr.2022.240863>.
- [11] A. Bradley, H. Barkema, A. Biggs, M. Green, T. Lam, Control of mastitis and enhancement of milk quality, in: Dairy Herd Health, CABI, Wallingford UK, 2012, pp. 117–168, <https://doi.org/10.1079/9781845939977.0117>.
 - [12] A. Bekuma, U. Galmessa, Review on hygienic milk products practice and occurrence of mastitis in cow's milk, Agric. Res. Technol. Open Access J. 18 (2) (2018) 1–11, <https://doi.org/10.19080/ARTOAJ.2018.18.556053>.
 - [13] M. Shoaib, A.I. Aqib, M.A. Naseer, Z.A. Bhutta, W. Pu, Q. Tanveer, M. Hamad, Etiology of Bovine Mastitis, IntechOpen, 2021, p. 67.
 - [14] A.K. Singh, A comprehensive review on subclinical mastitis in dairy animals: Pathogenesis, factors associated, prevalence, economic losses and management strategies, CABI Reviews (2022), <https://doi.org/10.1079/cabreviews202217057>, 2022.
 - [15] M. Rifatbegović, R.A. Nicholas, T. Mutevelić, M. Hadziomerović, Z. Maksimović, Pathogens associated with bovine mastitis: the experience of Bosnia and Herzegovina, Veterinary Sciences 11 (2) (2024) 63, <https://doi.org/10.3390/vetsci11020063>.
 - [16] A. Ashraf, M. Imran, Causes, types, etiological agents, prevalence, diagnosis, treatment, prevention, effects on human health and future aspects of bovine mastitis, Anim. Health Res. Rev. 21 (1) (2020) 36–49, <https://doi.org/10.1017/S1466252319000094>.
 - [17] J.A. Hertl, Y.T. Gröhn, J.D.G. Leach, D. Bar, G.J. Bennett, R.N. González, Y. H. Schukken, Effects of clinical mastitis caused by gram-positive and gram-negative bacteria and other organisms on the probability of conception in New York State Holstein dairy cows, J. Dairy Sci. 93 (4) (2010) 1551–1560, <https://doi.org/10.3168/jds.2009-2599>.
 - [18] K. Kibebew, Bovine mastitis: a review of causes and epidemiological point of view, Journal of Biology, Agriculture and Healthcare 7 (2) (2017) 1–14.
 - [19] L.K. Chung, S. Sahibzada, H.C. Annandale, I.D. Robertson, F.W. Waichigo, M. S. Tufail, J.A. Aleri, Bacterial pathogens associated with clinical and subclinical mastitis in a Mediterranean pasture-based dairy production system of Australia, Res. Vet. Sci. 141 (2021) 103–109, <https://doi.org/10.1016/j.rvsc.2021.10.005>.
 - [20] M.S. Kamel, N.M. Bakry, Clinical and subclinical mastitis, in: The Microbiology, Pathogenesis and Zoonosis of Milk Borne Diseases, Academic Press, 2024, pp. 153–190, <https://doi.org/10.1016/B978-0-443-13805-8.00006-5>.
 - [21] R. Meçaj, G. Muça, X. Koleci, M. Sulçe, L. Turmalaj, P. Zalla, M. Tafaj, Bovine environmental mastitis and their control: an overview, Int. J. Agric. Biosci. 12 (4) (2023) 216–221, <https://doi.org/10.47278/journal.ijab/2023.067>.
 - [22] A.H. Kober, S. Saha, M.A. Islam, M.S.R. Rajoka, K. Fukuyama, H. Aso, H. Kitazawa, Immunomodulatory effects of probiotics: a novel preventive approach for the control of bovine mastitis, Microorganisms 10 (11) (2022) 2255, <https://doi.org/10.3390/microorganisms10112255>.
 - [23] R. Paramasivam, D.R. Gopal, R. Dhandapani, R. Subbarayalu, M.P. Elangovan, B. Prabhu, S. Muthupandian, Is AMR in dairy products a threat to human health? An updated review on the origin, prevention, treatment, and economic impacts of subclinical mastitis, Infect. Drug Resist. (2023) 155–178.
 - [24] C. Sharma, N. Rokana, M. Chandra, B.P. Singh, R.D. Gulhane, J.P.S. Gill, H. Panwar, Antimicrobial resistance: its surveillance, impact, and alternative management strategies in dairy animals, Front. Vet. Sci. 4 (2018) 237, <https://doi.org/10.3389/fvets.2017.00237>.
 - [25] C. Bang, T. Dagan, P. Deines, N. Dubilier, W.J. Duschl, S. Fraune, T.C. Bosch, Metaorganisms in extreme environments: do microbes play a role in organismal adaptation? Zoology 127 (2018) 1–19, <https://doi.org/10.1016/j.zool.2018.02.004>.
 - [26] K. Sauer, P. Stoodley, D.M. Goeres, L. Hall-Stoodley, M. Burmölle, P.S. Stewart, T. Bjarnsholt, The biofilm life cycle: expanding the conceptual model of biofilm formation, Nat. Rev. Microbiol. 20 (10) (2022) 608–620.
 - [27] N. Kungwani, S.K. Shukla, T.S. Rao, S. Das, Biofilm-mediated bioremediation of polycyclic aromatic hydrocarbons: current status and future perspectives, Microbial Biodegradation and Bioremediation (2022) 547–570, <https://doi.org/10.1016/B978-0-323-85455-9.00021-7>.
 - [28] F. Nourbakhsh, M.S. Nasrollahzadeh, A.S. Tajani, V. Soheili, F. Hadizadeh, Bacterial biofilms and their resistance mechanisms: a brief look at treatment with natural agents, Folia Microbiol. 67 (4) (2022) 535–554, <https://doi.org/10.1007/s12223-022-00955-8>.
 - [29] A. Bekuma, U. Galmessa, Review on hygienic milk products practice and occurrence of mastitis in cow's milk, Agric. Res. Technol. Open Access J. 18 (2) (2018) 1–11, <https://doi.org/10.19080/artoaj.2016.02.555590>.
 - [30] L. Touza-Otero, M. Landin, P. Diaz-Rodriguez, Fighting antibiotic resistance in the local management of bovine mastitis, Biomed. Pharmacother. 170 (2024) 115967, <https://doi.org/10.1016/j.biopha.2023.115967>.
 - [31] S. Singh, J.P. Singh, R. Kumar, P. Kumar, Review on hygienic milk products practice and occurrence of mastitis in cow's milk, Agric. Res. Technol. Open Access J. 18 (2) (2024) 1–11, <https://doi.org/10.19080/ARTOAJ.2018.18.556053>. Mastitis. Periparturient Diseases of Cattle, 101–120. Bekuma, A., & Galmessa, U. (2018).
 - [32] B.D. Perry, D. Grace, K. Sones, Current drivers and future directions of global livestock disease dynamics, Proc. Natl. Acad. Sci. USA 110 (52) (2013) 20871–20877, doi/abs/10.1073/pnas.1012953108.
 - [33] H.C.J. Godfray, C. Donnelly, G. Hewinson, M. Winter, J. Wood, Bovine TB Strategy Review, 2018.
 - [34] P. Krishnamoorthy, K.P. Suresh, K.S. Jayamma, B.R. Shome, S.S. Patil, R. G. Amachawadi, An understanding of the global status of major bacterial pathogens of milk concerning bovine mastitis: a systematic review and meta-analysis (Scientometrics), Pathogens 10 (5) (2021) 545, <https://doi.org/10.3390/pathogens10050545>.
 - [35] J.L. Gonçalves, C. Kamphuis, C.M.M.R. Martins, J.R. Barreiro, T. Tomazi, A. H. Gameiro, M.V. Dos Santos, Bovine subclinical mastitis reduces milk yield and economic return, Livest. Sci. 210 (2018) 25–32, <https://doi.org/10.1016/j.livsci.2018.01.016>.
 - [36] A.K. Balhara, V. Nayan, A. Dey, K.P. Singh, S.S. Dahiya, I. Singh, Climate change and buffalo farming in major milk producing states of India—Present status and need for addressing concerns, Indian J. Anim. Sci. 87 (4) (2017) 403–411, <https://doi.org/10.56093/ijans.v87i4.69476>.
 - [37] K. Sharun, K. Dhama, R. Tiwari, M.B. Gungoo, M. Iqbal Yatoo, S.K. Patel, W. Chaicumpa, Advances in therapeutic and managerial approaches of bovine mastitis: a comprehensive review, Vet. Q. 41 (1) (2021) 107–136, doi/full/10.1080/01652176.2021.1882713.
 - [38] A. Haider, M. Ikram, I. Shahzadi, M. Asif Raza, Bovine mastitis, in: Polymeric Nanoparticles for Bovine Mastitis Treatment, Springer Nature Switzerland, Cham, 2023, pp. 49–80, https://doi.org/10.1007/978-3-031-39947-3_4.
 - [39] D. Bar, L.W. Tauer, G. Bennett, R.N. González, J.A. Hertl, H.F. Schulte, Y. T. Gröhn, Use of a dynamic programming model to estimate the value of clinical mastitis treatment and prevention options utilized by dairy producers, Agric. Syst. 99 (1) (2008) 6–12, <https://doi.org/10.1016/j.agsy.2008.09.001>. Aqib, A. I., Muneer, A., Shafeeq, M., &).
 - [40] N. Kirn, Economic impacts of clinical and sub clinical mastitis on dairy farms, Vet. Sci. Res. 3 (2) (2021) 31–39, <https://doi.org/10.30564/vsr.v3i2.4119>.
 - [41] A. Bansod, R. Masand, A. Jadhao, A. Bhardwaj, S. Singh, V. Gaikwad, An overview of subclinical mastitis in dairy cattle, Indian J. Ani Health (2021) 1–9, <https://doi.org/10.36062/ijah.2021.04521>.
 - [42] R. Patel, P. Kunjadia, P. Koringa, C. Joshi, A. Kunjadiya, Microbiological profiles in clinical and subclinical cases of mastitis in milking Jafarabadi buffalo, Res. Vet. Sci. 125 (2019) 94–99, <https://doi.org/10.1016/j.rvsc.2019.05.012>.
 - [43] V.M.A. Gallardo, Better understanding of dairy cow mastitis: incidence of clinical mastitis and pathogen-specific milk losses. Bedding Bacterial Counts, and E-Learning Training Methods, Cornell University, 2021.
 - [44] M. Asfaw, A. Negash, Review on impact of bovine mastitis in dairy production, Adv. Biol. Res. 11 (3) (2017) 126–131, <https://doi.org/10.5829/idosi.abr.2017.126.131>.
 - [45] H. Hiitö, J. Vakkamäki, H. Simojoki, T. Autio, J. Junnila, S. Pelkonen, S. Pyörälä, Prevalence of subclinical mastitis in Finnish dairy cows: changes during recent decades and impact of cow and herd factors, Acta Vet. Scand. 59 (2017) 1–14, <https://doi.org/10.1186/s13028-017-0288>.
 - [46] A.M. Heikkilä, E. Liski, S. Pyörälä, S. Taponen, Pathogen-specific production losses in bovine mastitis, J. Dairy Sci. 101 (10) (2018) 9493–9504, <https://doi.org/10.3168/jds.2018-14824>.
 - [47] P. Stanek, P. Żółkiewski, E. Januś, A review on mastitis in dairy cows research: current status and future perspectives, Agriculture 14 (8) (2024) 1292, <https://doi.org/10.3390/agriculture14081292>.
 - [48] J.W. Barlow, R.N. Zadoks, Y.H. Schukken, Effect of lactation therapy on Staphylococcus aureus transmission dynamics in two commercial dairy herds, BMC Vet. Res. 9 (2013) 1–12, <https://doi.org/10.1186/1746-6148-9-28>.
 - [49] S. Kour, N. Sharma, B. N. P. Kumar, J.S. Soodan, M.V.D. Santos, Y.O. Son, Advances in diagnostic approaches and therapeutic management in bovine mastitis, Veterinary Sciences 10 (7) (2023) 449, <https://doi.org/10.3390/vetsci10070449>.
 - [50] P. Sandhu, I. Kumari, S. Swargam, Proteomics in livestock health and diseases, Systems Biology, Bioinformatics and Livestock Science 167 (2023).
 - [51] A. Boix Amorós, Human Milk Microbiota and its Relationship with Milk Components in Health and during Lactational Mastitis (Doctoral Dissertation, Universitat Politècnica de València), 2019, <https://doi.org/10.4995/Thesis/10251/129870>.
 - [52] M. Kumar, A.K. Yadav, V. Verma, B. Singh, G. Mal, R. Nagpal, R. Hemalatha, Bioengineered probiotics as a new hope for health and diseases: an overview of potential and prospects, Future Microbiol. 11 (4) (2016) 585–600, <https://doi.org/10.2217/fmb.16.4>.
 - [53] A. Saleem, S.A. Saleem Bhat, F. Omonijo, N. A. Ganai, E.M. Ibeagha-Awemu, S. Mudasar Ahmad, Immunotherapy in mastitis: state of knowledge, research gaps and way forward, Vet. Q. 44 (1) (2024) 1–23, <https://doi.org/10.1080/01652176.2024.2363626>.
 - [54] J.C. Plaizier, M.D. Mesgaran, H. Derakhshani, H. Golder, E. Khafipour, J.L. Kleen, Q. Zebeli, Enhancing gastrointestinal health in dairy cows, Animal 12 (s2) (2018) s399–s418, <https://doi.org/10.1017/S1751731118001921>.
 - [55] M.I. Hossain, M.F.R. Mizan, M. Ashrafudoulla, S. Nahar, H.J. Joo, I.K. Jahid, S. D. Ha, Inhibitory effects of probiotic potential lactic acid bacteria isolated from kimchi against Listeria monocytogenes biofilm on lettuce, stainless-steel surfaces, and MBECTM biofilm device, Lwt 118 (2020) 108864, <https://doi.org/10.1016/j.lwt.2019.108864>.
 - [56] R.R. Pedersen, V. Krömker, T. Bjarnsholt, K. Dahl-Pedersen, R. Buhl, E. Jørgensen, Biofilm research in bovine mastitis, Front. Vet. Sci. 8 (2021) 656810, <https://doi.org/10.3389/fvets.2021.656810>.
 - [57] N. Berardo, L. Bohl, C. Porporatto, M.E.F. Nader-Macias, C. Bogni, M. Pellegrino, Intramammary inoculation with lactic acid bacteria at dry-off triggers an immunomodulatory response in dairy cows, Benef. Microbes 11 (6) (2020) 561–572, <https://doi.org/10.3920/BM2019.0163>.
 - [58] C.H. Jeong, W.N. Cheng, H. Bae, K.W. Lee, S.M. Han, M.C. Petriello, S.G. Han, Bee venom decreases LPS-induced inflammatory responses in bovine mammary epithelial cells, J. Microbiol. Biotechnol. 27 (10) (2017) 1827–1836, <https://doi.org/10.4014/jmb.1706.06003>.

- [59] C. Thirunavukkarasu, S.C. Watkins, C.R. Gandhi, Mechanisms of endotoxin-induced NO, IL-6, and TNF- α production in activated rat hepatic stellate cells: role of p38 MAPK, *Hepatology* 44 (2) (2006) 389–398, <https://doi.org/10.1002/hep.21254>.
- [60] G.T. Pinheiro Machado, M.B. Veleirinho, L. Mazzarino, L.C.P. Machado Filho, M. Maraschin, R.L.A. Cerri, S. Kuhnen, Development of propolis nanoparticles for the treatment of bovine mastitis: in vitro studies on antimicrobial and cytotoxic activities, *Can. J. Anim. Sci.* 99 (4) (2019) 713–723, <https://doi.org/10.1139/cjas-2018-0173>.
- [61] W.N. Cheng, S.G. Han, Bovine mastitis: risk factors, therapeutic strategies, and alternative treatments—a review, *Asian-Australasian journal of animal sciences* 33 (11) (2020) 1699, <https://doi.org/10.5713/ajas.20.0156>.
- [62] A.S. Neculai-Valeanu, A.M. Arton, Udder health monitoring for prevention of bovine mastitis and improvement of milk quality, *Bioengineering* 9 (11) (2022) 608, <https://doi.org/10.3390/bioengineering9110608>.
- [63] C. Nielsen, S. Østergaard, U. Emanuelson, H. Andersson, B. Berglund, E. Strandberg, Economic consequences of mastitis and withdrawal of milk with high somatic cell count in Swedish dairy herds, *Animal* 4 (10) (2010) 1758–1770, <https://doi.org/10.1017/S1751731110000704>.
- [64] P. Krishnamoorthy, K.P. Suresh, K.S. Jayamma, B.R. Shome, S.S. Patil, R. G. Amachawadi, An understanding of the global status of major bacterial pathogens of milk concerning bovine mastitis: a systematic review and meta-analysis (scientometrics), *Pathogens* 10 (5) (2021) 545, <https://doi.org/10.3390/pathogens10050545>.
- [65] A.E. Nura, B.A. Fitwi, E.G. Leykun, Factors associated with bacteriological cure of *S. aureus* in bovine mastitis in mid rift valley of Ethiopia, <http://meddocsonline.org/>, 2021.
- [66] T.J. McGenitty, A. Gessesse, J.E. Hallsworth, E. Garcia Cela, C. Verheeeke-Vaessen, F. Wang, K. Timmis, Visualizing the invisible: class excursions to ignite children's enthusiasm for microbes, *Biotechnol.* 13 (4) (2020) 844–887, <https://doi.org/10.1111/1751-7915.13576>.
- [67] M. Kumar, D.K. Sarma, S. Shubham, M. Kumawat, V. Verma, P.B. Nina, R. R. Tiwari, Futuristic non-antibiotic therapies to combat antibiotic resistance: a review, *Front. Microbiol.* 12 (2021) 609459, <https://doi.org/10.3389/fmicb.2021.609459>.
- [68] C. Carrascosa, D. Raheem, F. Ramos, A. Saraiva, A. Raposo, Microbial biofilms in the food industry—a comprehensive review, *Int. J. Environ. Res. Publ. Health* 18 (4) (2021) 2014, <https://doi.org/10.3390/ijerph18042014>.
- [69] F.L. Nobrega, A.R. Costa, L.D. Kluskens, J. Azeredo, Revisiting phage therapy: new applications for old resources, *Trends Microbiol.* 23 (4) (2015) 185–191, <https://doi.org/10.1016/j.tim.2015.01.006>.
- [70] A. Kakasis, G. Panitsa, Bacteriophage therapy as an alternative treatment for human infections. A comprehensive review, *Int. J. Antimicrob. Agents* 53 (1) (2019) 16–21, <https://doi.org/10.1016/j.ijantimicag.2018.09.004>.
- [71] K. Sharun, K. Dhama, R. Tiwari, M.B. Gugjoo, M. Iqbal Yatoo, S.K. Patel, W. Chaicumpa, Advances in therapeutic and management approaches of bovine mastitis: a comprehensive review, *Vet. Q.* 41 (1) (2021) 107–136, <https://doi.org/10.1080/01652176.2021.1882713>.
- [72] H. Haddad Kashani, M. Schmelcher, H. Sabzalipoor, E. Seyed Hosseini, R. Moniri, Recombinant endolysins as potential therapeutics against antibiotic-resistant *Staphylococcus aureus*: current status of research and novel delivery strategies, *Clin. Microbiol. Rev.* 31 (1) (2018) 10–1128, <https://doi.org/10.1128/cmr.00071-17>.
- [73] P. Rainard, F.B. Gilbert, P. Germon, G. Foucras, Invited review: a critical appraisal of mastitis vaccines for dairy cows, *J. Dairy Sci.* 104 (10) (2021) 10427–10448, <https://doi.org/10.3168/jds.2021-20434>.
- [74] W.N. Cheng, S.G. Han, Bovine mastitis: risk factors, therapeutic strategies, and alternative treatments—a review, *Asian-Australasian journal of animal sciences* 33 (11) (2020) 1699, <https://doi.org/10.5713/ajas.20.0156>.
- [75] J. Tiwari, C. Babra, H. Tiwari, V. Williams, S. De Wet, J. Gibson, T. Mukkur, Trends in therapeutic and prevention strategies for management of bovine mastitis: an overview, *J. Vaccines Vaccin.* 4 (1) (2013) 1–11, <https://doi.org/10.4172/2157-7560.1000176>.
- [76] K. Garber, Drugging the gut microbiome: using conventional drug discovery and novel synthetic biology approaches, some investigators and companies are mining our resident microbes and their metabolites for targets in small-molecule drug programs, *Nat. Biotechnol.* 33 (3) (2015) 228–232.
- [77] D. Zamojska, A. Nowak, I. Nowak, E. Macierzyńska-Piotrowska, Probiotics and postbiotics as substitutes of antibiotics in farm animals: a review, *Animals: an Open Access Journal From MDPI* 11 (12) (2021) 3431, <https://doi.org/10.3390/ani1123431>.
- [78] Y. Wang, X. Nan, Y. Zhao, L. Jiang, H. Wang, F. Zhang, B. Xiong, Dietary supplementation of inulin ameliorates subclinical mastitis via regulation of rumen microbial community and metabolites in dairy cows, *Microbiol. Spectr.* 9 (2) (2021) e00105–e00121, <https://doi.org/10.1128/Spectrum.00105-21>.
- [79] T. Tatsaporn, K. Kornkanok, Using potential lactic acid bacteria biofilms and their compounds to control biofilms of foodborne pathogens, *Biotechnology Reports* 26 (2020) e00477, <https://doi.org/10.1016/j.btre.2020.e00477>.
- [80] A. Angelopoulou, A.K. Warda, C. Hill, R.P. Ross, Non-antibiotic microbial solutions for bovine mastitis—live biotherapeutics, bacteriophage, and phage lysins, *Crit. Rev. Microbiol.* 45 (5–6) (2019) 564–580, <https://doi.org/10.1080/1040841X.2019.1648381>.
- [81] N. Berardo, L. Bohl, C. Porporatto, M.E.F. Nader-Macias, C. Boggi, M. Pellegrino, Intramammary inoculation with lactic acid bacteria at dry-off triggers an immunomodulatory response in dairy cows, *Benef. Microbes* 11 (6) (2020) 561–572, <https://doi.org/10.3920/BM2019.0163>.
- [82] R.F. Leite, J.L. Gonçalves, A.P.F. Peti, F.S. Figueiró, L.A.B.D. Moraes, M.V. D. Santos, Antimicrobial activity of crude extracts from actinomycetes against mastitis pathogens, *J. Dairy Sci.* 101 (11) (2018) 10116–10125, <https://doi.org/10.3168/jds.2018-14454>.
- [83] R. Bava, F. Castagna, V. Musella, C. Lupia, E. Palma, D. Britti, Therapeutic Use of Bee venom and potential applications in veterinary medicine, *Veterinary Sciences* 10 (2) (2023) 119, <https://doi.org/10.3390/vetsci10020119>.
- [84] S.C. Cheng, W.C. Huang, J.H.S. Pang, Y.H. Wu, C.Y. Cheng, Quercetin inhibits the production of IL-1 β -induced inflammatory cytokines and chemokines in ARPE-19 cells via the MAPK and NF- κ B signaling pathways, *Int. J. Mol. Sci.* 20 (12) (2019) 2957, <https://doi.org/10.3390/ijms20122957>.
- [85] D.G. Téllez-de-Jesús, N.S. Flores-Lopez, J.A. Cervantes-Chávez, A.R. Hernández-Martínez, Antibacterial and antifungal activities of encapsulated Au and Ag nanoparticles synthesized using Argemone mexicana L extract, against antibiotic-resistant bacteria and *Candida albicans*, *Surface. Interfac.* 27 (2021) 101456, <https://doi.org/10.1016/j.surfint.2021.101456>.
- [86] E.M.M. Abubakar, Efficacy of crude extracts of garlic (*Allium sativum* Linn.) against nosocomial *Escherichia coli*, *Staphylococcus aureus*, *Streptococcus pneumoniae* and *Pseudomonas aeruginosa*, *J. Med. Plants Res.* 3 (4) (2009) 179–185. ISSN 1996-0875 © 2009 Academic Journals.
- [87] A.K. Singh, R.K. Sharma, V. Sharma, T. Singh, R. Kumar, D. Kumari, Isolation, morphological identification and in vitro antibacterial activity of endophytic bacteria isolated from *Azadirachta indica* (neem) leaves, *Vet. World* 10 (5) (2017) 510, <https://doi.org/10.14202/vetworld.2017.510-516>.
- [88] F.R. Mendes, F.G. Silva, E.O. Sousa, F.F. Rodrigues, J.G. Costa, F.J. Monte, J. C. Assunção, Essential oil of *Alpinia zerumbet* (Pers.) BL Burtt. & RM Sm. (Zingiberaceae): chemical composition and modulation of the activity of aminoglycoside antibiotics, *J. Essent. Oil Res.* 27 (3) (2015) 259–263, <https://doi.org/10.1080/10412905.2015.1014935>.
- [89] I.N. Piccinin, A.A. Zielinski, S. Kuhnen, Invasive plant *Bidens pilosa* as an ecofriendly antibiofilm-antimicrobial against *Staphylococcus aureus* for bovine mastitis control, *Organic Agriculture* 13 (1) (2023) 73–82, <https://doi.org/10.1007/s13165-022-00415-0>.
- [90] A.O. Gaspar, C.C. Guizelini, F.C. Roberto, G.S. Difante, R.C. Brumatti, C.C. Ítavo, S.T. Lee, Protodioscin levels in *Brachiaria* spp. in a sheep production system and a brief review of the literature of *Brachiaria* spp. poisoning in ruminants, *Pesqui. Vet. Bras.* 41 (2021) e06921, <https://doi.org/10.1590/1678-5150-PVB-6921>.
- [91] M.E. Light, L.J. McGaw, S.G. Sparg, A.K. Jäger, J. Van Staden, Screening of *Cenchrus ciliaris* L. for biological activity, *South Afr. J. Bot.* 68 (3) (2002) 411–413, [https://doi.org/10.1016/S0254-6299\(15\)30410-5](https://doi.org/10.1016/S0254-6299(15)30410-5).
- [92] P. Sakthar, B. Chauhan, Antibacterial, antioxidant and cell proliferative properties of *Coccinia grandis* fruits, *Avicenna journal of phytomedicine* 7 (4) (2017) 295.
- [93] N. Ergün, G. Ökmen, P. Erdal, Z. Canteekin, Y. Ergün, The antibacterial activities of *Lavandula stoechas* and *Crepis sancta* leaf and flower against mastitis pathogens and enzymatic and non-enzymatic antioxidant activities of the extracts, *Turkish Journal of Agriculture-Food Science and Technology* 6 (5) (2018) 543–549, <https://doi.org/10.24925/turjaf.v6i5.543-549.1692>.
- [94] A. Karnwal, In vitro antibacterial activity of *Hibiscus rosa sinensis*, *Chrysanthemum indicum*, and *Calendula officinalis* flower extracts against Gram negative and Gram positive food poisoning bacteria, *Advances in Traditional Medicine* 22 (3) (2022) 607–619, [https://doi.org/10.1016/S2221-1691\(12\)60064-1](https://doi.org/10.1016/S2221-1691(12)60064-1).
- [95] N. Singh, S. Gupta, V. Rathore, Comparative antimicrobial study of ethanolic extract of leaf and rhizome of *Curcuma longa* Linn, *Phcog. J.* 9 (2) (2017), <https://doi.org/10.5530/pj.2017.2.35>.
- [96] M.I. Naik, B.A. Fomda, E. Jaykumar, J.A. Bhat, Antibacterial activity of lemongrass (*Cymbopogon citratus*) oil against some selected pathogenic bacteria, *Asian Pac. J. Tropical Med.* 3 (7) (2010) 535–538, [https://doi.org/10.1016/S1995-7645\(10\)60129-0](https://doi.org/10.1016/S1995-7645(10)60129-0).
- [97] L.S. Varghese, M.A. Ninan, N.E.E.N.A. Alex, S. Soman, J. Simi, Comparative antibacterial activity of fruit extracts of *Embellica officinalis* Gaertn. Against gram positive versus gram negative bacteria, *Biomedical & Pharmacology Journal* 6 (2) (2013) 447, <https://doi.org/10.13005/bpj/441>.
- [98] I.A. Reshi, T.K. Sarkar, A.H. Bhat, H.U. Malik, A. Muhee, S. Qureshi, Antibacterial activity of some medicinal plants of Kashmir, *J&K, India, J. Pure Appl. Microbiol.* 11 (1) (2017) 521–527, <https://doi.org/10.22207/JPAM.11.1.68>.
- [99] B. Vasiljević, D. Mitić-Čulafić, I. Djekić, T. Marković, J. Knežević-Vukčević, I. Tomasević, B. Nikolić, Antibacterial effect of *Juniperus communis* and *Satureja Montana* essential oils against *Listeria monocytogenes* in vitro and in wine marinated beef, *Food Control* 100 (2019) 247–256, <https://doi.org/10.1016/j.foodcont.2019.01.025>.
- [100] A.E. Al-Snafi, H.S. Khadem, H.A. Al-Saedy, A.M. Alqahtani, G. El-Saber, A review on *Medicago sativa*: a potential medicinal plant, *Int. J. Biol. Pharm. Sci. Arch* 1 (2021) 22–33, <https://doi.org/10.30574/ijbps.2021.1.2.0302>.
- [101] K.A. Hadid, A.K. Shakya, R.R. Naik, N. Jaradat, H.S. Farah, N. Shalan, G. A. Oriqat, Phenolic content and antioxidant and antimicrobial activities of *Malva sylvestris* L., *Malva oxyloba* Boiss., *Malva parviflora* L., and *Malva aegyptia* L. leaves extract, *J. Chem.* 2021 (2021) 1–10, <https://doi.org/10.1155/2021/8867400>.
- [102] R.A. Hassan, S.T. Abotaleb, H.B. Hamed, M.S. Eldeen, Antioxidant and biological activities of *mentha longifolia*, L. Extracts, *Journal of Agricultural Chemistry and Biotechnology* 11 (3) (2020) 59–62, <https://doi.org/10.21608/jac.2020.86996>.
- [103] N.P. Raut, T.A. Shafi, M.P. Sakhare, M.F.M.F. Siddiqui, S.S. Gaikward, G. D. Ranvir, Therapeutic evaluation of *Murraya koenigii* in bovine subclinical

- mastitis, *J. Pharmacogn. Phytochem.* 10 (4S) (2021) 40–45, <https://doi.org/10.22271/phyto.2021.v10.i4Sa.14288>.
- [104] M.V. Moliva, L.N. Cariddi, E.R. Pereyra, J.M. Raviolo, N. Sambuceti, A. Posadaz, E.B. Reinoso, Evaluation of antibacterial and antibiofilm properties of *Minthostachys verticillata* essential oils against bovine *Staphylococcus aureus* strains, *Biocatal. Agric. Biotechnol.* 50 (2023) 102697, <https://doi.org/10.1016/j.bcab.2023.102697> Getrightsandcontent.
- [105] B. Ashrafi, P. Ramak, B. Ezatpour, G.R. Talei, Biological activity and chemical composition of the essential oil of *Nepeta cataria* L, *J Res Pharm* 23 (2) (2019) 336–343, <https://doi.org/10.12991/jrp.2019.141>.
- [106] J.P. Singh, S. Singh, D.K. Yadav, D. Niyogi, D.P. Shrivastava, R.K. Joshi, Antimicrobial efficacy of *Nyctanthes arbor-tristis* Linn. Leaf extract against common pathogens of sub-clinical mastitis in cows, *Indian Journal of Veterinary Sciences & Biotechnology* 18 (3) (2022) 16–20.
- [107] M. Pebam, M.V. Sushma, S.A. Sankaranarayanan, A.M. Thanekar, N. Koyande, A. K. Rengan, Antiviral perspectives of economically important Indian medicinal plants and spices, *Proceedings of the Indian National Science Academy* 88 (3) (2022) 392–416, <https://doi.org/10.3389/fphar.2021.732891>.
- [108] M. Gherib, B. Chahrazed, I.A. El-Haci, T.M. Chaouche, F.A. Bekkara, Antioxidant and antibacterial activities of aerial part essential oil and some organic extracts from the Algerian medicinal plant *Pulicaria mauritanica* Cass., *Int. J. Pharmaceut. Sci. Res.* 7 (1) (2016) 76, [https://doi.org/10.13040/IJPSR.0975-8232.7\(1\).76-84](https://doi.org/10.13040/IJPSR.0975-8232.7(1).76-84).
- [109] S. Kumar, V. Singh, V. Johar, S. Varma, R. Thakur, Antioxidant activity, antibacterial activity and phenolic composition of pomegranate fruit: a review, *J Food Chem Nanotechnol* 9 (S1) (2023) S588–S593, <https://doi.org/10.17756/jfcn.2023-s1-074>.
- [110] S. Darmayani, S. Alaydrus, R. Yunus, T. Yuniarty, N.P. Dewi, A. Rosanty, Antibacterial activities test of ethanol extracts of kundur fruit (*benincasa hispida* thunb. Cogn) on *Salmonella typhi* bacteria, *J. Phys. Conf.* 1899 (1) (2021, May) 012028, <https://doi.org/10.1088/1742-6596/1899/1/012028>. IOP Publishing.
- [111] G.K. Padhy, *Spathodea campanulata* P. Beauv.—a review of its ethnomedicinal, phytochemical, and pharmacological profile, *J. Appl. Pharmaceut. Sci.* 11 (12) (2021) 17–44, <https://doi.org/10.7324/JAPS.2021.1101202>.
- [112] J.B. Perianayagam, S.K. Sharma, K.K. Pillai, A. Pandurangan, D. Kesavan, Evaluation of antimicrobial activity of ethanol extract and compounds isolated from *Trichodesma indicum* (Linn.) R. Br. root, *J. Ethnopharmacol.* 142 (1) (2012) 283–286, <https://doi.org/10.1016/j.jep.2012.04.020>.
- [113] Bera, A., & Banerjee, N. Estimation of Total Phenolic and Flavonoid content, Antibacterial and Antioxidant Potential of *Tridax procumbens* Linn. from Paschim Medinipur. *In Vitro*, 29, 0-430. <https://www.researchgate.net/publication/372100959>.
- [114] W.N. Cheng, S.G. Han, Bovine mastitis: risk factors, therapeutic strategies, and alternative treatments—a review, *Asian-Australasian journal of animal sciences* 33 (11) (2020) 1699, <https://doi.org/10.5713/ajas.20.0156>.
- [115] A.J. Martín-Rodríguez, H. Quezada, G. Becerril, C. Fuente-Núñez, I. De Castillo-Juarez, Recent advances in novel antibacterial development, *Front. Clin. Drug Res. Anti Infect* 2 (2016) 3–61, <https://doi.org/10.4103/2229-516X.112229>.
- [116] L. Xu, Y.Y. Wang, J. Huang, C.Y. Chen, Z.X. Wang, H. Xie, Silver nanoparticles: synthesis, medical applications and biosafety, *Theranostics* 10 (20) (2020) 8996, <https://doi.org/10.7150/thno.45413>.
- [117] Z. Yu, Q. Li, J. Wang, Y. Yu, Y. Wang, Q. Zhou, P. Li, Reactive oxygen species-related nanoparticle toxicity in the biomedical field, *Nanoscale Res. Lett.* 15 (1) (2020) 115, <https://doi.org/10.1186/s11671-020-03344-7>.
- [118] J.A. Hernández-Díaz, J.J. Garza-García, A. Zamudio-Ojeda, J.M. León-Morales, J. C. López-Velázquez, S. García-Morales, Plant-mediated synthesis of nanoparticles and their antimicrobial activity against phytopathogens, *J. Sci. Food Agric.* 101 (4) (2021) 1270–1287, <https://doi.org/10.1002/jsfa.10767>.
- [119] A. Kalińska, S. Jaworski, M. Wierzbicki, M. Gołębiewski, Silver and copper nanoparticles—an alternative in future mastitis treatment and prevention? *Int. J. Mol. Sci.* 20 (7) (2019) 1672, <https://doi.org/10.3390/ijms20071672>.
- [120] A. Gibała, P. Żeliszewska, T. Gosiewski, A. Krawczyk, D. Duraczyńska, J. Szaleniec, M. Oćwieja, Antibacterial and antifungal properties of silver nanoparticles—effect of a surface-stabilizing agent, *Biomolecules* 11 (10) (2021) 1481, <https://doi.org/10.3390/biom11101481>.
- [121] A.K. Mohammed, K.K. Salh, F.A. Ali, ZnO, TiO₂ and Ag nanoparticles impact against some species of pathogenic bacteria and yeast, *Cellular and Molecular Biology* 67 (3) (2021) 24–34, <https://doi.org/10.14715/cmb/2021.67.3.4>.
- [122] A. Elbehry, M. Al-Dubaib, E. Marzouk, I. Moussa, Antibacterial effects and resistance induction of silver and gold nanoparticles against *Staphylococcus aureus*-induced mastitis and the potential toxicity in rats, *Microbiology Open* 8 (4) (2019) e00698, <https://doi.org/10.1002/mbo3.698>.
- [123] Y.G. Yuan, Q.L. Peng, S. Gurunathan, Effects of silver nanoparticles on multiple drug-resistant strains of *Staphylococcus aureus* and *Pseudomonas aeruginosa* from mastitis-infected goats: an alternative approach for antimicrobial therapy, *Int. J. Mol. Sci.* 18 (3) (2017) 569, <https://doi.org/10.3390/ijms18030569>.
- [124] K. Azarin, A. Usatov, T. Minkina, A. Plotnikov, A. Kasyanova, A. Fedorenko, S. Alamri, Effects of ZnO nanoparticles and its bulk form on growth, antioxidant defense system and expression of oxidative stress related genes in *Hordeum vulgare* L, *Chemosphere* 287 (2022) 132167, <https://doi.org/10.1016/j.chemosphere.2021.132167>.
- [125] M. De Barros, P.G. Perciano, M.H. Dos Santos, L.L. De Oliveira, É.D.M. Costa, M.A. S. Moreira, Antibacterial activity of 7-Epiclusianone and its novel copper metal complex on *Streptococcus* spp. isolated from bovine mastitis and their cytotoxicity in MAC-T Cells, *Molecules* 22 (5) (2017) 823, <https://doi.org/10.3390/molecules22050823>.
- [126] B. Prusa, Alternative solutions to antibiotics in mastitis treatment for dairy cows—a review, *Animal Sci Papers Rep* 38 (2) (2020) 117–133, <https://doi.org/10.3389/fvets.2023.1160350>.
- [127] C. Sharma, J. M Al Kaabi, S. M Nurulain, S. N Goyal, M. Amjad Kamal, S. Ojha, Polypharmacological properties and therapeutic potential of β -caryophyllene: a dietary phytocannabinoid of pharmaceutical promise, *Curr. Pharmaceut. Des.* 22 (21) (2016) 3237–3264, <https://doi.org/10.2174/138161282266616031115226>.
- [128] C. Sandoval-Salinas, J.P. Saffon, J.M. Martínez, H.A. Corredor, A. Gallego, Are radial pressure waves effective for the treatment of moderate or mild to moderate erectile dysfunction? A randomized sham therapy controlled clinical trial, *J. Sex. Med.* 19 (5) (2022) 738–744, <https://doi.org/10.1016/j.jsexm.2022.02.010>.